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Latihan I

A. Menggunakan teorema dasar integral tentu !

3) $\int_1^2 (x+1)^2 \cdot dx = \left[\frac{1}{3}(x+1)^3 \right]_1^2$

$= \left[\frac{1}{3}(2+1)^3 \right] - \left[\frac{1}{3}(1+1)^3 \right]$

$= \frac{27}{3} - \frac{8}{3}$

$= \frac{19}{3}$

4) $\int_1^4 (t^2 - t - 6) \cdot dt = \frac{1}{3}t^3 - \frac{1}{2}t^2 - 6t \Big]_1^4$

$= \left(\frac{1}{3}(4)^3 - \frac{1}{2}(4)^2 - 6(4) \right) - \left(\frac{1}{3} - \frac{1}{2} - 6 \right)$

$= \left(\frac{64}{3} - 32 \right) - \left(\frac{2 - 3 - 36}{6} \right)$

$= \frac{64 - 96}{3} + \frac{37}{6}$

$= \frac{128 - 192 + 37}{6}$

$= -\frac{27}{6} = -4.5$

B. Integral tentu fungsi trigonometri

1) $\int_{\pi/3}^{\pi/2} 3 \sin(2\theta) \cdot d\theta = 3 \int_{\pi/3}^{\pi/2} \sin(2\theta) d\theta$

$= 3 \cdot \frac{1}{2} \left[-\cos(2\theta) \right]_{\pi/3}^{\pi/2}$

$= -\frac{3}{2} \cos(2\theta) \Big]_{\pi/3}^{\pi/2}$

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$$= \left(-\frac{3}{2}\cos\left(2\cdot\frac{\pi}{2}\right)\right) - \left(-\frac{3}{2}\cos\left(2\cdot\frac{\pi}{3}\right)\right)$$

$$= \left(-\frac{3}{2}\cos(180^\circ)\right) - \left(-\frac{3}{2}\cos(120^\circ)\right)$$

$$= \left(-\frac{3}{2}(-1)\right) - \left(-\frac{3}{2}\left(-\frac{1}{2}\right)\right)$$

$$= \frac{3}{2} - \frac{3}{4}$$

$$= \frac{6-3}{4}$$

$$= \frac{3}{4} = 0{,}75$$

2) $\int_0^{\pi/2} \sin(2x-\pi)\,dx = \left[-\frac{1}{2}\cos(2x-\pi)\right]_0^{\pi/2}$

$$= \left[-\frac{1}{2}\cos\left(2\left(\frac{\pi}{2}\right)-\pi\right)\right] - \left[-\frac{1}{2}\cos(2(0)-\pi)\right]$$

$$= \left[-\frac{1}{2}\cos(0^\circ)\right] - \left[-\frac{1}{2}\cos(-180^\circ)\right]$$

$$= \left(-\frac{1}{2}(1)\right) - \left(-\frac{1}{2}(-1)\right)$$

$$= -\frac{1}{2} - \frac{1}{2}$$

$$= -1$$

C. Hitung integral berikut untuk t antara 0 dan 2π (juga antara $-\pi$ & π)

1) $\int_{2\pi}^{0} \cos t\,dt = \sin t \Big]_{2\pi}^{0}$

$= \sin(0) - \sin(2\pi)$

$= 0 - 0$

$= 0$

$\int_{\pi}^{-\pi} \cos t\,dt = \sin t \Big]_{\pi}^{-\pi}$

$= \sin(-\pi) - \sin(\pi)$

$= 0 - 0$

$= 0$

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5) $\int_{2\pi}^{0} \sin 2t \cos 3t \cdot dt = \int_{2\pi}^{0} \frac{1}{2}\left(\sin(5t) + \sin(-t)\right) \cdot dt$

$= \int_{2\pi}^{0} \frac{1}{2}\left(\sin(5t) - \sin(t)\right) \cdot dt$

$= \frac{1}{2}\int_{2\pi}^{0} \left(\sin(5t) - \sin(t)\right) \cdot dt$

$= \frac{1}{2}\left[-\frac{1}{5}\cos(5t) + \cos(t)\right] + C$

$= \left[-\frac{1}{10}\cos(5t) + \frac{1}{2}\cos(t)\right]_{2\pi}^{0}$

$= \left(-\frac{1}{10}\cos(0) + \frac{1}{2}\cos(0)\right) - \left(-\frac{1}{10}\cos(10\pi) + \frac{1}{2}\cos(2\pi)\right)$

$= \left(-\frac{1}{10}(1) + \frac{1}{2}\cos(1)\right) - \left(-\frac{1}{10}(1) + \frac{1}{2}(1)\right)$

$= \frac{4}{10} - \frac{4}{10}$

$= 0$

$\int_{\pi}^{-\pi} \sin 2t \cos 3t \cdot dt = \int_{\pi}^{-\pi} \frac{1}{2}\left(\sin(5t) - \sin(t)\right) \cdot dt$

$= \frac{1}{2}\left[-\frac{1}{5}\cos(5t) + \cos(t)\right]_{\pi}^{-\pi}$

$= \left[-\frac{1}{10}\cos(5t) + \frac{1}{2}\cos(t)\right]_{\pi}^{-\pi}$

$= \left(-\frac{1}{10}\cos(-5\pi) + \frac{1}{2}\cos(-\pi)\right) - \left(-\frac{1}{10}\cos(\pi) + \frac{1}{2}\cos(\pi)\right)$

$= \left(-\frac{1}{10}(-1) + \frac{1}{2}(-1)\right) - \left(-\frac{1}{10}(-1) + \frac{1}{2}(-1)\right)$

$= 0$

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Latihan 2

Hitung luas daerah yang dibatasi oleh kurva y dan batas - batas x yang diberikan !

1. $y = 3x^2$ antara $x = 1$ dan $x = 2$

$$A = \int_1^2 3x^2 \cdot dx = \left[x^3 \right]_1^2$$
$$= (2)^3 - (1)^3$$
$$= 8 - 1$$
$$= 7 \text{ satuan luas}$$

7. $y = x^3 - 4x^2 + 4x$ antara $x = 0$ dan $x = 2$

$$A = \int_0^2 (x^3 - 4x^2 + 4x) \cdot dx = \left[\frac{1}{4}x^4 - \frac{4}{3}x^3 + 2x^2 \right]_0^2$$
$$= \left[\frac{1}{4}(2)^4 - \frac{4}{3}(2)^3 + 2(2)^2 \right] - 0$$
$$= \frac{16}{4} - \frac{32}{3} + 8$$
$$= 4 - \frac{32}{3} + 8$$
$$= 12 - \frac{32}{3}$$
$$= \frac{36}{3} - \frac{32}{3}$$
$$= \frac{4}{3}$$
$$= 1\frac{1}{3} \text{ satuan luas}$$

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Latihan 3

A. Gambarkan grafik fungsi berikut, dan hitung nilai rata-rata dan nilai efektif f !

1. $f(x) = 9 - x^2 \; ; (-3, 3)$

$$f_{average} = \frac{\int_{-3}^{3} (9 - x^2) \, dx}{3 - (-3)}$$

$$= \frac{\left[9x - \frac{1}{3}x^3\right]_{-3}^{3}}{6}$$

$$= \frac{\left(9(3) - \frac{1}{3}(3)^3\right) - \left(9(-3) - \frac{1}{3}(-3)^3\right)}{6}$$

$$= \frac{\left(27 - \frac{27}{3}\right) - \left((-27) + \frac{27}{3}\right)}{6}$$

$$= \frac{(27 - 9) - (-27 + 9)}{6}$$

$$= \frac{36}{6}$$

$$= 6$$

$$f_{rms} = \sqrt{\frac{1}{b-a} \int_{-3}^{3} (9 - x^2)^2 \cdot dx}$$

$$= \sqrt{\frac{1}{3-(-3)} \int_{-3}^{3} (81 - 18x^2 + x^4) \cdot dx}$$

$$= \sqrt{\frac{1}{6} \left[81x - 6x^3 + \frac{1}{5}x^5\right]_{-3}^{3}}$$

$$= \sqrt{\frac{1}{6} \left[\left(81(3) - 6(3)^3 + \frac{1}{5}(3)^5\right) - \left(81(-3) - 6(-3)^3 + \frac{1}{5}(-3)^5\right)\right]}$$

$$= \sqrt{\frac{1}{6} \left(\left(243 - 162 + \frac{243}{5}\right) - \left(-243 + 162 - \frac{243}{5}\right)\right)}$$

$$= \sqrt{\frac{1}{6} \left(\left(\frac{1215 - 810 + 243}{5}\right) + \frac{1215 - 810 + 243}{5}\right)}$$

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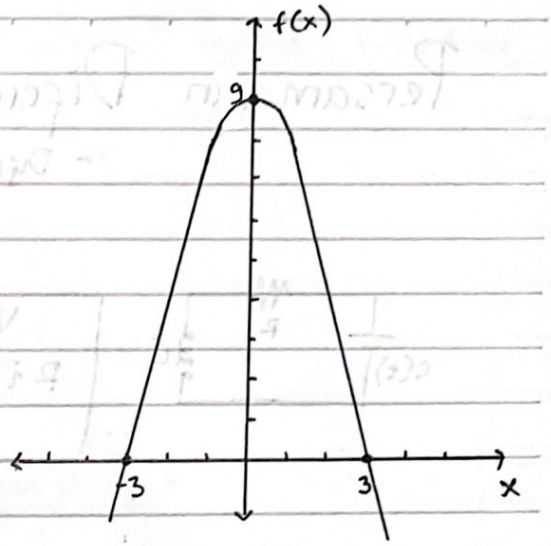
$$= \sqrt{\frac{1}{6}\left(\frac{648}{5} + \frac{648}{5}\right)}$$

$$= \sqrt{\frac{1}{6}\left(\frac{1296}{5}\right)}$$

$$= \sqrt{\frac{1296}{30}}$$

$$= \frac{36}{\sqrt{30}}$$

$$= 6.57$$



2. $f(x) = x \quad , \quad 0 \leq x \leq \pi$

$$f_{average} = \frac{\int_0^{\pi} x \cdot dx}{\pi - 0}$$

$$= \frac{\frac{1}{2}x^2 \Big|_0^{\pi}}{\pi}$$

$$= \frac{\frac{1}{2}(\pi)^2}{\pi}$$

$$= \frac{\pi^2}{2\pi}$$

$$= \frac{\pi}{2}$$

$$f_{rms} = \sqrt{\frac{1}{b-a}\int_0^{\pi} f^2(x) \cdot dx}$$

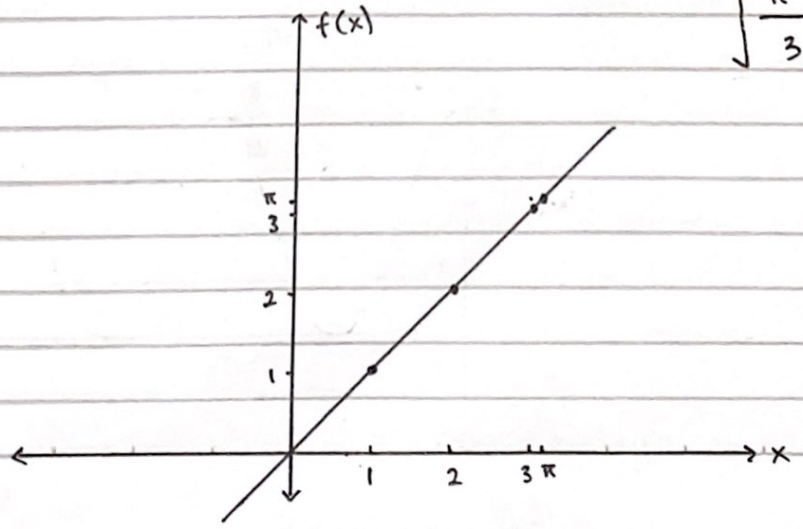
$$= \sqrt{\frac{1}{\pi - 0}\int_0^{\pi} x^2 \cdot dx}$$

$$= \sqrt{\frac{1}{\pi}\left(\frac{1}{3}x^3\right)_0^{\pi}}$$

$$= \sqrt{\frac{1}{\pi}\left(\frac{\pi^3}{3}\right)}$$

$$= \sqrt{\frac{\pi^3}{3\pi}}$$

$$= \sqrt{\frac{\pi^2}{3}} = \frac{\pi}{\sqrt{3}}$$



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B. Gambarkan grafik fungsi periodik dengan periode T berikut, kemudian hitung nilai rata - rata dan nilai efektifnya !

6. $f(t) = 3\cos^2 t \Rightarrow T = \frac{2\pi}{1} = 2\pi$

$$f_{average} = \frac{\int_0^{2\pi} (3\cos^2 t).dt}{2\pi - 0}$$

$$= \frac{\int_0^{2\pi} (3 \cdot \cos t \cdot \cos t).dt}{2\pi}$$

$$= \frac{3\int_0^{2\pi} \left(\frac{1}{2}(\cos 2t + \cos 0)\right).dt}{2\pi}$$

$$= \frac{3 \cdot \frac{1}{2}\int_0^{2\pi} (\cos(2t) + 1).dt}{2\pi}$$

$$= \frac{\frac{3}{2}\left[\frac{1}{2}\sin 2t + t\right]_0^{2\pi}}{2\pi}$$

$$= \frac{\frac{3}{2}\left(\left(\frac{1}{2}\sin 2(2\pi) + 2\pi\right) + 0\right)}{2\pi}$$

$$= \frac{\frac{3}{2}\left(\frac{1}{2}\sin 4\pi + 2\pi\right)}{2\pi}$$

$$= \frac{\frac{3}{4}\sin 4\pi + 3\pi}{2\pi}$$

$$= \frac{\frac{3}{4}(0) + 3\pi}{2\pi}$$

$$= \frac{3\pi}{2\pi}$$

$$= \frac{3}{2}$$

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$$f_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi}(3\cos^2 t)^2 \cdot dt}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi}(9\cos^4 t)\cdot dt}$$

$$= \sqrt{\frac{9}{2\pi}\int_0^{2\pi}(\cos^2 t)^2 \cdot dt}$$

$$= \sqrt{\frac{9}{2\pi}\int_0^{2\pi}\left(\frac{1}{2}\left(1+\cos(2t)\right)\right)^2 \cdot dt}$$

$$= \sqrt{\frac{9}{2\pi}\int_0^{2\pi}\frac{1}{4}\left(1+2\cos(2t)+\cos^2(2t)\right)\cdot dt}$$

$$= \sqrt{\frac{9}{2\pi}\cdot\frac{1}{4}\left[t+\sin 2t+\left(\frac{1}{2}\left(x+\frac{1}{4}\sin(4t)\right)\right)\right]_0^{2\pi}}$$

$$= \sqrt{\frac{9}{8\pi}\left[t+\sin(2t)+\frac{1}{2}x+\frac{1}{8}\sin(4t)\right]_0^{2\pi}}$$

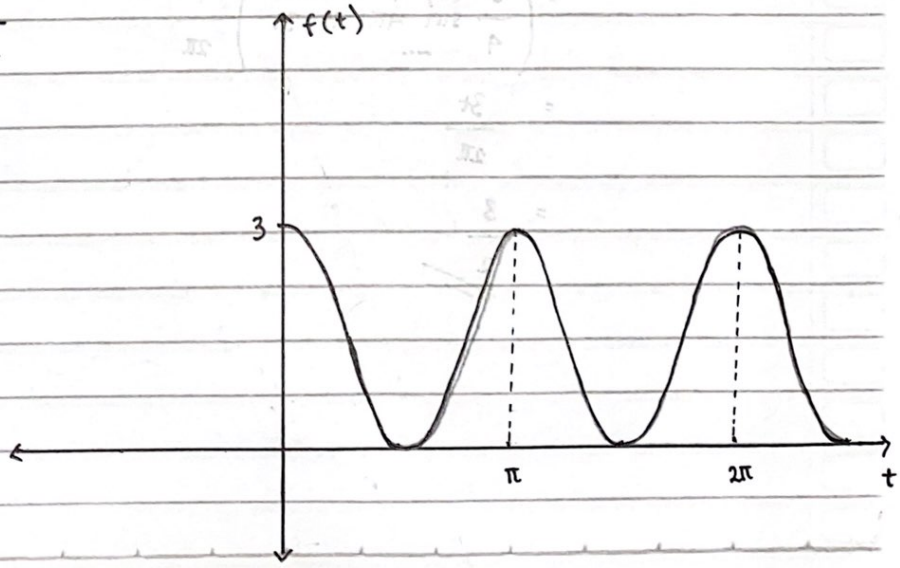
$$= \sqrt{\frac{9}{8\pi}\left(2\pi+\sin(4\pi)+\frac{2\pi}{2}+\frac{\sin(8\pi)}{8}\right)}$$

$$= \sqrt{\frac{9}{8\pi}\left(3\pi+0+0\right)}$$

$$= \sqrt{\frac{27\pi}{8\pi}}$$

$$= \frac{3}{2}\sqrt{\frac{3}{2}}$$

$$= \frac{3\sqrt{6}}{4}$$ //



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10. $f(x) = \begin{cases} x+2 & \text{bila } -\pi < x < 0 \\ -x+2 & \text{bila } 0 < x < \pi \end{cases}$

dengan $T = 2\pi$

$$f_{average} = \frac{1}{2\pi}\left[\int_{-\pi}^{0}(x+2)\,dx + \int_{0}^{\pi}(-x+2)\,dx\right]$$

$$= \frac{1}{2\pi}\left[\left(\frac{1}{2}x^2+2x\right)_{-\pi}^{0} + \left(-\frac{1}{2}x^2+2x\right)_{0}^{\pi}\right]$$

$$= \frac{1}{2\pi}\left[\left(0-\left(\frac{\pi^2}{2}-2\pi\right)\right) + \left(-\frac{\pi^2}{2}+2\pi\right)\right]$$

$$= \frac{1}{2\pi}\left[-\frac{\pi^2}{2}+2\pi-\frac{\pi^2}{2}+2\pi\right]$$

$$= \frac{1}{2\pi}\left[-\frac{2\pi^2}{2}+4\pi\right]$$

$$= \frac{-\pi^2+4\pi}{2\pi}$$

$$f_{rms} = \sqrt{\frac{1}{2\pi}\left[\int_{-\pi}^{0}(x+2)^2\,dx + \int_{0}^{\pi}(-x+2)^2\,dx\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[\int_{-\pi}^{0}(x^2+4x+4)\,dx + \int_{0}^{\pi}(x^2-4x+4)\,dx\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[\left(\frac{1}{3}x^3+2x^2+4x\right)_{-\pi}^{0} + \left(\frac{1}{3}x^3-2x^2+4x\right)_{0}^{\pi}\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[0-\left(-\frac{\pi^3}{3}+2\pi^2-4\pi\right)+\frac{\pi^3}{3}-2\pi^2+4\pi\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[\frac{\pi^3}{3}-2\pi^2+4\pi+\frac{\pi^3}{3}-2\pi^2+4\pi\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[2\frac{\pi^3}{3}-4\pi^2+8\pi\right]}$$

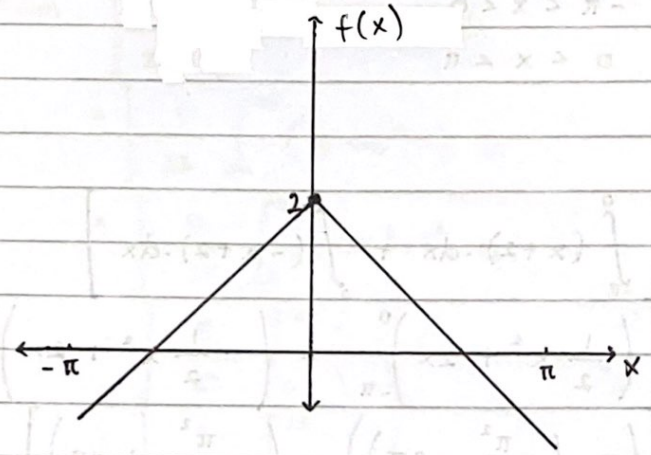
$$= \sqrt{\frac{2\pi^3}{6\pi}-\frac{4\pi^2}{2\pi}+\frac{8\pi}{2\pi}}$$

$$= \sqrt{\frac{\pi^2}{3}-2\pi+4}$$

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